

TITLE PAGE

Problem Statement ID	PS 4
Problem Statement Title	AI-powered behavioral anomaly detection for cybersecurity
Theme	Smart Automation / Cybersecurity
PS Category	Software
Student Name	Bhagya Paras Morabia
Student ID	20527297
IDEA TITLE	Sentinel Flow: Hybrid Multi-Layered UEBA System

IDEA TITLE: SENTINEL FLOW: HYBRID MULTI-LAYERED UEBA SYSTEM

Detailed explanation of the proposed solution:

- **Production-Grade UEBA:** Sentinel Flow monitors enterprise access logs to detect compromised accounts and insider threats.
- **AI-Driven Profiling:** Replaces rule-heavy SIEMs by learning mathematically "normal" behaviors (geo-location, resource access, temporal habits) for every individual network entity.

How it addresses the problem:

- **Alert Fatigue Reduction:** Fusing multiple models achieves an incredible 99.71% Precision @ Top 1% alert budget, ensuring SOC analysts only see highly validated, actionable threats.
- **Handles Dynamic Environments:** Automatically adapts to changing user behaviors (Concept Drift) using ADWIN algorithms, and handles brand new employees (Cold-Start) using EWMA cohort fallbacks.

Innovation and uniqueness of the solution:

- **Tri-Layer Architecture:** We do not rely on a single "black box" model. We run a Deterministic Rule Engine, an Unsupervised Isolation Forest, and a Sequence-Aware Markov model in parallel.
- **Explainable AI (XAI):** Anomalies aren't just flagged; the XGBoost Fusion layer uses SHAP values to explain exactly why a risk score is high, mapping the behavior directly to the MITRE ATT&CK framework.

TECHNICAL APPROACH

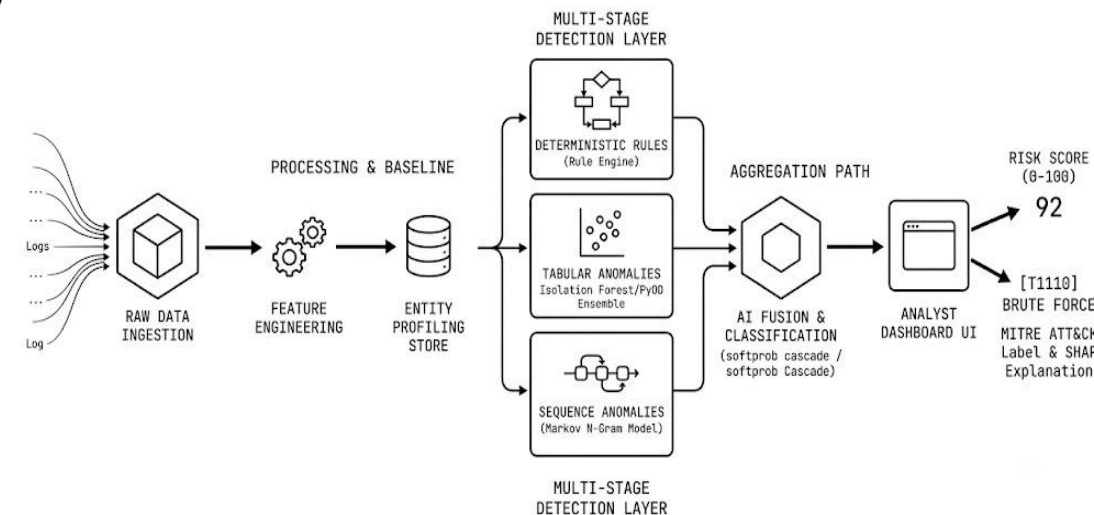
Technologies to be used:

- **Backend ML & API:** Python, FastAPI (Asynchronous ASGI Microservice), Pandas/NumPy.
- **Machine Learning:** XGBoost (Fusion Classifier), Scikit-Learn (Isolation Forest / ECOD).
- **Frontend Dashboard:** Vanilla JavaScript, HTML5, CSS (Tailwind-inspired utility classes), DOM-Hydration for zero-latency updates.

Methodology and process:

- **Phase 1 (Ingestion & Profiling):** High-throughput FastAPI endpoints asynchronously ingest log data and extract behavioral features (geo-velocity, resource entropy, temporal shifts).
- **Phase 2 (Parallel Detection):** Data is routed through three distinct models simultaneously:
 - **Rule Engine:** Catches impossible travel and brute force mathematically.
 - **Tabular Model:** Detects statistical outliers (e.g., massive data exfiltration).
 - **Sequence Model:** Tracks chronological context to catch lateral movement.
- **Phase 3 (Supervised Fusion):** Outputs are fed into an XGBoost classifier which acts as the final judge, outputting a calibrated 0-100 Risk Score and specific attack classification.

SENTINELFLOW PROJECT ARCHITECTURE FLOW



FEASIBILITY AND VIABILITY

Analysis of feasibility:

- **Real-Time Execution:** Production-tested libraries (PyOD, XGBoost) enable sub-15ms event inference.
- **Scalability:** Decoupled microservice architecture allows independent backend and frontend scaling for enterprise volumes.

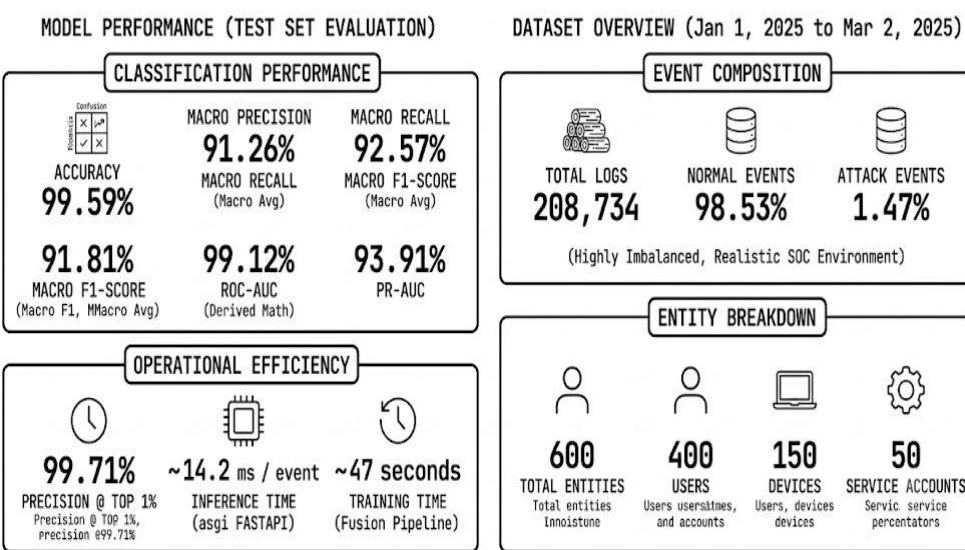
Potential challenges & risks:

- **Alert Fatigue:** Natural, legitimate changes in user behavior (e.g., traveling) triggering false positive alerts.
- **Cold-Start Problem:** Accurately scoring new employees or devices that lack historical baseline data.

Strategies for overcoming challenges:

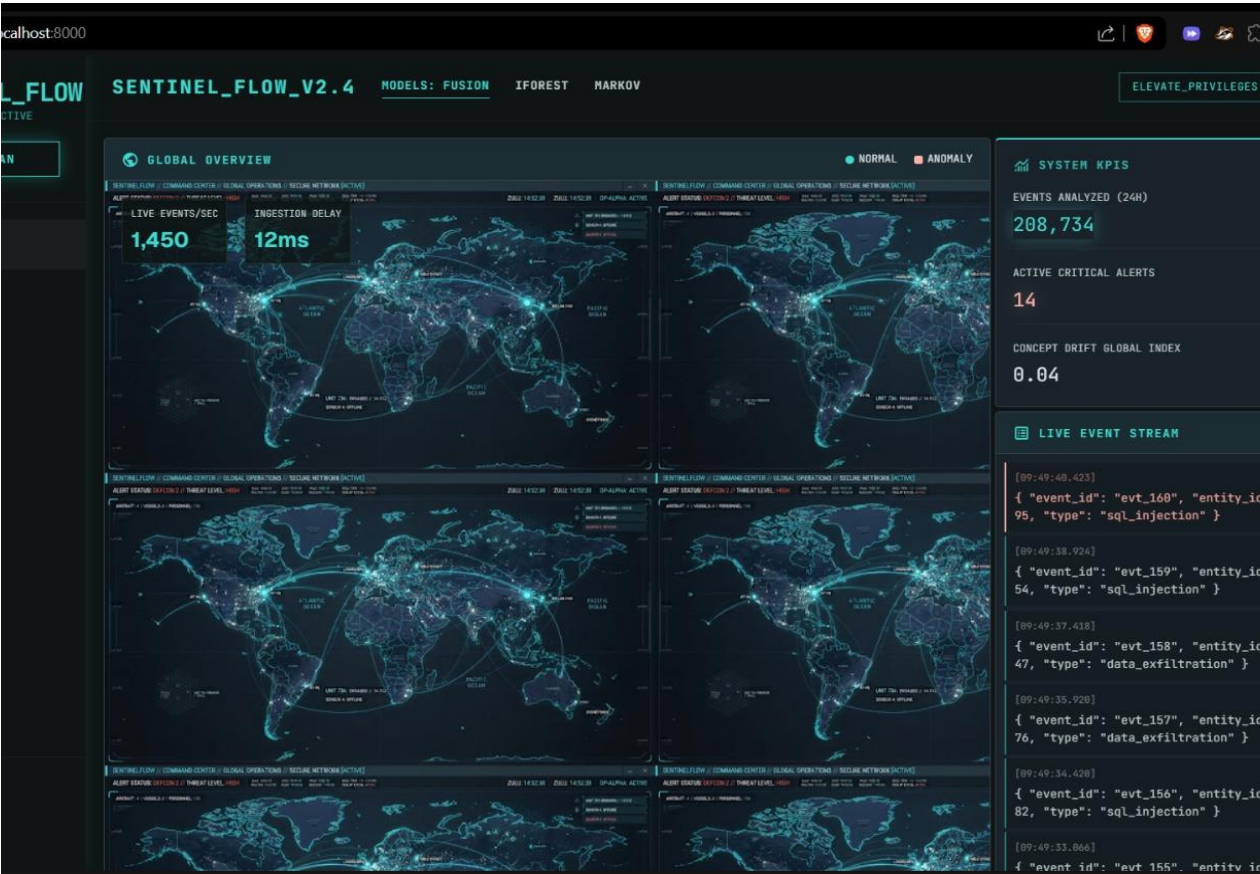
- **Concept Drift Management:** Implemented River's ADWIN algorithm to track statistical shifts and automatically widen acceptance windows for users changing roles.
- **Cohort Blending (EWMA):** New entities temporarily inherit peer group baselines, decaying to their own empirical data over time.

SENTINELFLOW PROJECT: PERFORMANCE METRICS & DATASET STATISTICS SUMMARY



ARTIFACTS

Relevant artifacts (Dashboard snaps):



Dashboard : Global Overview & Main Screen



Entity Profile: Behavioral Baselines & Concept Drift Tracker

RESEARCH AND REFERENCES

Details / Links of the reference and research work:

- Du, Li, Zheng, Srikumar. DeepLog: Anomaly Detection and Diagnosis from System Logs through Deep Learning. ACM CCS 2017. <https://doi.org/10.1145/3133956.3134015> (PDF: <https://users.cs.utah.edu/~lifeifei/papers/deeplog.pdf>)
- Chen, Guestrin. XGBoost: A Scalable Tree Boosting System. KDD 2016. <https://arxiv.org/abs/1603.02754>
- River — online machine learning library, ADWIN drift detector: <https://riverml.xyz> and <https://riverml.xyz/dev/api/drift/ADWIN/>
- Masquerading / device spoofing (T1036): <https://attack.mitre.org/techniques/T1036/>
- MITRE ATT&CK framework home: <https://attack.mitre.org>
- Li, Zhao, Hu, Botta, Ionescu, Chen. ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions. IEEE TKDE 2022. <https://doi.org/10.1109/TKDE.2022.3159580>
- PyOD library (Isolation Forest, ECOD, and 50+ other detectors, unified API): <https://github.com/yzhao062/pyod>